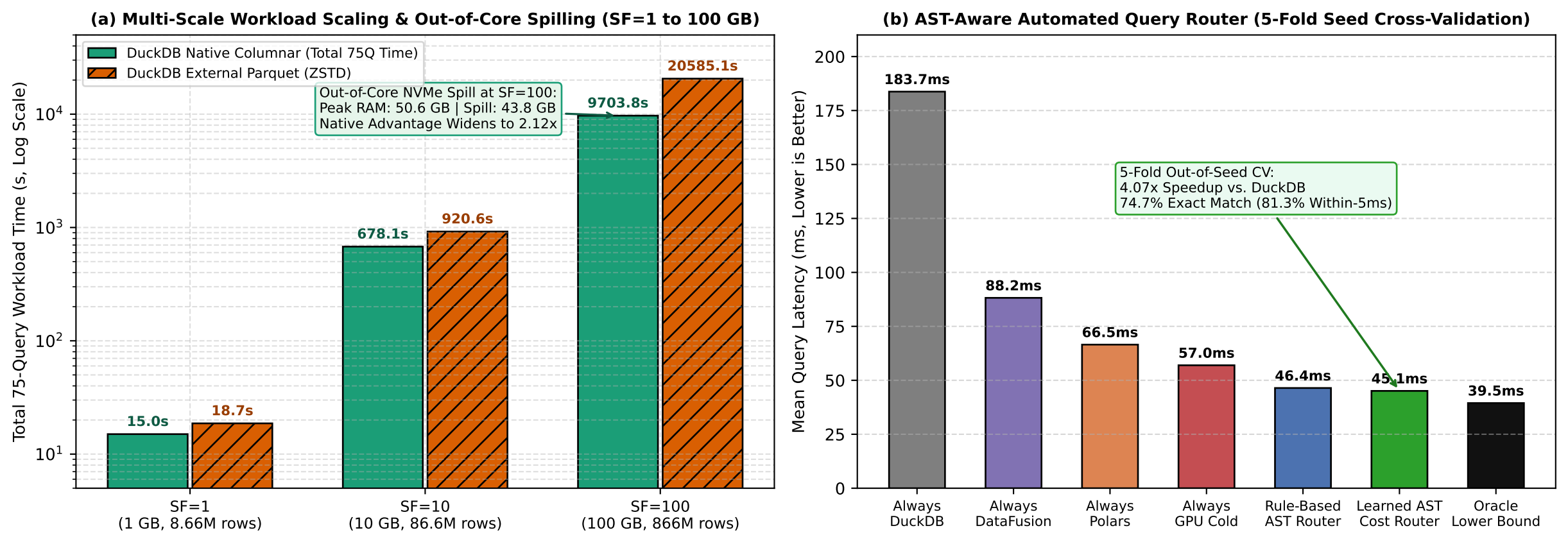
(a) Multi-Scale Workload Scaling & Out-of-Core Spilling (SF=1 to 100 GB)
DuckDB Native Columnar (Total 75Q Time)
DuckDB External Parquet (ZSTD)
Out-of-Core NVMe Spill at SF=100:
Peak RAM: 50.6 GB | Spill: 43.8 GB
Native Advantage Widens to 2.12x
Total 75-Query Workload Time (s, Log Scale)
10^4
10^3
10^2
10^1
15.0s
18.7s
678.1s
920.6s
9703.8s
20585.1s
SF=1
(1 GB, 8.66M rows)
SF=10
(10 GB, 86.6M rows)
SF=100
(100 GB, 866M rows)
(b) AST-Aware Automated Query Router (5-Fold Seed Cross-Validation)
Mean Query Latency (ms, Lower is Better)
200
175
150
125
100
75
50
25
0
5-Fold Out-of-Seed CV:
4.07x Speedup vs. DuckDB
74.7% Exact Match (81.3% Within-5ms)
183.7ms
88.2ms
66.5ms
57.0ms
46.4ms
45.1ms
39.5ms
Always
DuckDB
Always
DataFusion
Always
Polars
Always
GPU Cold
Rule-Based
AST Router
Learned AST
Cost Router
Oracle
Lower Bound